

Erika C. Freeman

Junior Group Leader (Tenure Track), IGB Berlin

erika.freeman@igb-berlin.de · ORCID 0000-0001-7161-6038 · Google Scholar E6WubOUAAAAJ · erikacfreeman.com · Swiss & Canadian citizen · English (native), German (fluent, B2–C1), French (basic), Spanish (basic)

Current position

Junior Group Leader (Tenure Track) — group *Aquatic Biogeochemistry of Carbon* (ABC Lab), Leibniz-Institute of Freshwater Ecology and Inland Fisheries (IGB), Department 1 (Ecohydrology & Biogeochemistry); Programme Area 4. Since **January 2026**. Research programme: the *ecology of molecules* — dissolved organic matter (DOM) at molecular resolution (FT-ICR-MS) to predict and manage carbon-climate feedbacks and water quality.

Appointments

Role	Institution	Dates
Postdoctoral Researcher (with F. Altermatt)	Eawag (ETH Domain) & University of Zürich	2023–2025
Remote Team Facilitator, WHONDRS Consortium	PNNL (international, ~20 countries)	2021–present
Program Coordinator, NSERC CREATE ABATE	(5-country)	2016–2018
Team Coordinator, Boreal 2050	Cdn. Network for Aquatic Ecosystem Services	2016–2018
Research Assistant	Western University	2013–2016

Education

Degree	Institution	Date	Supervisor(s)
PhD, Plant Sciences	University of Cambridge	defended Apr 2023	A. Tanentzap; E. Emilson (Canadian Forest Service)
MSc, Geography / Global Health Systems	Western University	2018	I. Creed
BSc, Environmental Science	Western University	2016	

Research funding & analytical awards

- **DOE EMSL CAM #62138** (*Molecular Fingerprints of Metal Fate*), sole PI — **approved Feb 2026**; free PNNL FT-ICR-MS / NMR / multi-omics access.
- **DOE EMSL FICUS #62211** (*AI-Enabled Phytomining of Critical Minerals*), co-PI with N. Jafari (Texas A&M) — **invited to full proposal, Mar 2026**.
- **IGB start-up budget** €50,000.
- *Applications in preparation*: DFG Sachbeihilfe TRACE (Sept 2026); DFG Emmy Noether / FreshDreaMS; ERC Starting Grant WATERMARK (2027); Leibniz Junior Research Group VITALS (Mar 2028).

Honours & awards

- **Gates Cambridge Scholarship** (2018–2023; ~1.3% acceptance rate).
- UTSC Postdoctoral Fellowship (2022–2024, declined).
- NSERC Canada Graduate Scholarship (M); Ontario Graduate Scholarship; NSERC USRA (2014, 2016); Cambridge fieldwork & travel awards.
- *Science* “Working Life” essay (2024), among the year’s most-read; re-featured for International Women’s Day 2025.

Publications

Full record below. First author: **8 published**, 4 in review or submitted. Co-author: **9 published**. Metrics via ORCID 0000-0001-7161-6038 and Google Scholar (E6WubOUAAAAJ).

First-author

- Freeman EC, Emilson EJS, Webster K, Dittmar T, Tanentzap AJ. Logging Disrupts the Ecology of Molecules in Headwater Streams. *PNAS* (2025).
- Freeman EC, McClure EA, Mudunuru ML, Ward CS, Bottos EM, González-Pinzón R, Feeser KL, Krause S, Peña J, Newcomer ME. Molecular Diversity of Dissolved Organic Matter Reflects Macroecological Patterns in River Networks. *Scientific Reports* (2025).

- Freeman EC, Peller T, Altermatt F. Ecosystem ecology needs an ecology of molecules. *Trends in Ecology and Evolution* (2025).
- Freeman EC, Padilla-Iglesias C. Racing the Clock. *Science* (2024).
- Freeman EC, Erratt KJ. Cyanobacteria in the Anthropocene: synanthropism forged in an era of global change. *Water Environmental Research* (2024).
- Freeman EC, Emilson EJS, Braga L, Emilson C, Martineau C, Dittmar T, Goldhammer T, Singer G, Tanentzap AJ. Universal Microbial Reworking of Dissolved Organic Matter along Environmental Gradients. *Nature Communications* (2024).
- Freeman EC, Jones B, Bergström AK, Creed IF. Global Changes May Be Promoting a Rise in Select Cyanobacteria in Nutrient-poor Northern Lakes. *Global Change Biology* (2020).
- Freeman EC, Erdozain M, Ouellet Dallaire C, Teichert S, Nelson HW, Creed IF. Demand for provisioning ecosystem services as a driver of change in the Canadian boreal zone. *Environmental Reviews* (2019).
- Freeman EC, Peller T, Keck F, Altermatt F. The Ecology of Molecules. *American Naturalist*, revised (R1); decision pending.
- Freeman EC. Rare Molecules as Community Architects. *Trends in Ecology and Evolution*, minor revision; acceptance pending.
- Freeman EC, Altermatt F. When molecules matter for ecosystem management. *Frontiers in Ecology and the Environment*, under review.
- Freeman EC. Reading the room at ASLO-SIL 2026: 1461 talks and the shape of a field. *L&O Bulletin*, submitted (peer-reviewed; not on IGB list).

Contributing-author

- Matula E, Emilson EJS, Smenderovac E, Fonvielle J, **Freeman EC**, Tanentzap AJ. Wildfires Change Summertime Dissolved Organic Matter in Boreal Headwater Streams. *JGR Biogeosciences* (2025).
- Erratt KJ, Creed IF, **Freeman EC**, Trick C. Missing the Middle: Deep Cyanobacteria Layers Pose a “Known Unknown” Risk to Water Security. *Water Security* (2023).
- Stadler M, de Melo M, Garayburu-Caruso V, **Freeman EC**, Bice K, Mateus-Barros E, Lu Y, Barnard M, Cheng S, Dwivedi D, Tanentzap A, Meile C. Applying the Core-Satellite Species Concept: Characteristics of Rare and Common Riverine Dissolved Organic Matter. *Frontiers in Water* (2023).
- Buser-Young J, Garcia P, Schrenk M, Regier P, Ward ND, Biçe K, Brooks S, **Freeman EC**, Lønborg C. Determining the Biogeochemical Transformations of Organic Matter Composition in Global Rivers Using Molecular Signatures. *Frontiers in Water* (2023).
- Erratt KJ, Creed IF, **Freeman EC**, Westrick J, Trick CG, Birbeck J, Watson L, Zastepa A. Deep Cyanobacteria Layers: An Overlooked Aspect of Managing Risks of Cyanobacteria. *Environmental Science & Technology* (2022).
- Borton MA, Collins SM, Graham EB, Garayburu-Caruso VA, Goldman AE, de Melo M, **Freeman EC**, & WHONDRS Crowdsourced Consortium. It Takes a Village: Using a Crowdsourced Approach to Investigate Organic Matter Composition in Global Rivers Through the Lens of Ecological Theory. *Frontiers in Water* (2022).
- Deglint JL, Park JA, **Freeman EC**, Erratt KJ, Allaf MM, Wong A. Feasibility of Instance Segmentation of Phytoplankton Using Brightfield Microscopy & Deep Learning. *Journal of Computational Vision and Imaging Systems* (2021).
- Lembrechts J, Van den Hoogen J, Aalto J, Ashcroft M, De Frenne P, Kemppinen J, **Freeman EC**, & Hoffrén R. Global Maps of Soil Temperature. *Global Change Biology* (2021).
- Creed IF, Bergström AK, Trick CG, Grimm NB, Hessen DO, Karlsson J, Kidd KA, Kritzberg E, McKnight DM, **Freeman EC**, Senar OE, & others. Global Change-driven Effects on Dissolved Organic Matter Composition: Implications for Food Webs of Northern Lakes. *Global Change Biology* (2018).
- Creed IF, Trick CG, **Freeman EC**, Oakes B. Protocol for the analysis of phytoplankton using the FlowCAM® from Fluid Imaging Technologies. Standard Operating Procedure 001-2016 (2016).

Protocols

Publishing the group’s DOM methods openly on protocols.io (sampling and preservation, FT-ICR-MS PPL-SPE workflow, AI-assisted molecular annotation); DOIs added as released.

Invited talks (selected)

- *If more UHR DOM data is the answer, what was the question?* – ASLO-SIL 2026, Montreal.
- PLANET-DOM workshop, UFZ Leipzig (Sep 2025).
- Complex Organic Mixtures in Aquatic Systems (COMPASS), Delmenhorst (Mar 2025).
- BEEES seminar, University of Zürich (Mar 2025).
- *Synthesising DOM Across Ecosystems*, Skiathos (Oct 2023).

Teaching & mentoring

- **Supervised** five Cambridge research projects / theses (G. Mezciems, B. Tan, J. Gayford, L. Mahony, W. Pallier) and two field technicians (K. Waxenberg, J. Schadenburg).
- **Teaching:** Teaching Assistant, Western University (2016–2018); guest lecture, FU Berlin freshwater-ecology seminar (Spring 2026).

Service & esteem

- **Coordinator**, WHONDRS Consortium (PNNL; ~20 countries).

- **Reviewer:** *Science, Science Advances, Nature Communications, Limnology & Oceanography, Geochimica et Cosmochimica Acta, Communications Earth & Environment, Biogeosciences.*
- **Grant / fellowship review:** DFG (individual research grants); NSERC Discovery; GWIS National Fellowship.
- **Session chair:** ASLO 2027 (San Juan; with J. Wang); a DFG meeting session.
- Cross-IGB roles: co-leads the predictive-ecology working group; contributes to the Cross-cutting Theme/Platform (CTP) Data Management.

Skills

Python, R, Google Earth Engine, bash, hydrological modeling (WhiteboxTools); statistics (GLM, mixed-effects, Bayesian, multivariate, machine learning); FT-ICR-MS / Orbitrap; AI annotation pipeline; geospatial analysis.